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ONE-SHOT CIRCUIT WITH CURRENT COMPENSATION

Abstract

A circuit includes a current compensation circuit having first and second compensation inputs, and a compensation output. The first compensation input is coupled to an input voltage terminal and the second compensation input is coupled to a switching terminal. The circuit further includes a one-shot circuit having a comparator having first and second comparator inputs, and a comparator output. The first comparator input is coupled to a voltage reference terminal and the second comparator input is coupled to the compensation output. The one-shot circuit further includes a switch coupled between the second comparator input and a ground terminal and a capacitor having first and second capacitor terminals, in which the first capacitor terminal is coupled to the second comparator input and the second capacitor terminal is coupled to the ground terminal.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of prior U.S. application Ser. No. 18/312,562 filed May 4, 2023, the entirety of which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This description relates to electrical circuits, in particular, to circuits and systems implementing a one-shot circuit with current compensation.

BACKGROUND

[0003] Power electronic systems are used in numerous devices. For example, portable electronic devices such as cellphones and laptops include a power electronic component called the converter that converts an input power level to an output power level based on a power requirement for a sub-circuit of the portable electronic device. Due to the vast number of sub-circuits that are present in these complex devices, there is a need to cater to varied voltage requirements that are not the same as the battery or the AC supply that powers the device.

[0004] In many electronic devices, the DC current signal applied to a circuit can have a residual AC portion. The AC portion is referred to as the ripple current. The ripple current can lead to increased heating and power dissipation in electronic components. Accordingly, efforts are made to keep ripple current low.

SUMMARY

[0005] An example circuit includes a current compensation circuit having first and second compensation inputs, and a compensation output, in which the first compensation input is coupled to an input voltage terminal and the second compensation input is coupled to a switching terminal. The circuit further includes a one-shot circuit including a comparator having first and second comparator inputs, and a comparator output, in which the first comparator input is coupled to a voltage reference terminal and the second comparator input is coupled to the compensation output. The one-shot circuit further includes a switch coupled between the second comparator input and a ground terminal. The one-shot circuit also includes a capacitor coupled between the second comparator input and the ground terminal.

[0006] Another example circuit includes a current compensation circuit configured to provide a compensated current based on a first voltage and a second voltage, wherein the first voltage is proportional to an input voltage and the second voltage is proportional to an average value of a switching voltage. The circuit further includes a one-shot circuit including a capacitor configured to provide a capacitor voltage based on the compensated current, a comparator configured to provide a comparator signal based on the capacitor voltage and a reference voltage, and a switch configured to discharge the capacitor.

[0007] As another example, a system including a converter that includes a high-side switch having first and second high-side terminals, in which the first high-side terminal is coupled to an input voltage terminal and the second high-side terminal is coupled to a switching terminal. The converter further includes a low-side switch having first and second low-side terminals, in which the first low-side terminal is coupled to the switching terminal and the second low-side terminal is coupled to a ground terminal, and an inductor having first and second inductor terminals, in which the first inductor terminal is coupled to the switching terminal and the second inductor terminal is coupled to an output terminal. The system further includes a current compensation circuit having first and second compensation inputs, and a compensation output, in which the first compensation input is coupled to the input voltage terminal and the second compensation input is coupled to the

switching terminal. Furthermore, the system includes a one-shot circuit including a comparator having first and second comparator inputs, and a comparator output, in which the first comparator input is coupled to a voltage reference terminal and the second comparator input is coupled to the compensation output. The one-shot circuit further includes a switch coupled between the second comparator input and the ground terminal; and a capacitor having first and second capacitor terminals, in which the first capacitor terminal is coupled to the second comparator input and the second capacitor terminal is coupled to the ground terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates a block diagram of an example circuit.

[0009] FIG. 2 is an example system that includes the circuit of FIG. 1.

[0010] FIG. 3 illustrates an example voltage sensor circuit.

[0011] FIG. 4 illustrates an example system that includes a current compensation circuit and a one-shot circuit.

[0012] FIG. 5 is a signal diagram that depicts the comparator output signal, the shot signal and a switching voltage of FIG. 2, for example.

DETAILED DESCRIPTION

[0013] This description relates to circuits and systems that implement a one-shot circuit with current compensation, such as can be implemented in power converter modules to provide constant inductor current ripple.

[0014] Power converter circuits employ high-frequency power conversion using switches and other passive components like inductors and capacitors to get a desired output voltage. With the need to meet the rapidly shrinking product footprints, power electronic systems are using higher and higher switching frequencies, to take advantage of smaller inductors and capacitors. However, using smaller inductors and capacitors leads to increased input ripple current and noise. For example, the power converter circuits are implemented as power converter modules, for example, DC-DC modules, to be utilized as part of compact electronic devices. In such modules, the value of the inductor is fixed. In other words, the value of the inductor cannot be varied in accordance with the output voltage to be generated. As a result, when the output voltage increases, the inductor ripple current increases. However, the inductor ripple current values have to be limited to meet the system requirements, and as a result only low values of output voltage can be supported by the power converter modules, thereby limiting their performance.

[0015] In an example, a circuit includes a current compensation circuit and a one-shot circuit. The one-shot circuit includes a capacitor and a comparator. The comparator is configured to provide an output signal to turn off a high-side switch of the power converter circuit. The current compensation circuit is configured to provide a compensated current, which is proportional to a difference between an input voltage and output voltage of a power converter circuit. The compensated current charges the capacitor of the one-shot circuit and the comparator is configured to turn off the high-side switch of the power converter circuit responsive to the voltage across the capacitor reaching a reference voltage. Therefore, the output of the comparator of the one-shot circuit controls an on-time of the high-side switch of the power converter circuit. The on-time of the high-side switch refers to a continuous time period during which the high-side switch is ON. Controlling the on-time of the high-side switch of the power converter circuit based on constant values of the one-shot circuit, as described below, helps to achieve a constant inductor ripple current in the power converter circuit. As a result, the circuit (e.g., a system on chip (SOC) module), which includes the current compensation circuit and the one-shot circuit, can provide higher output voltages without increasing the inductor ripple current.

[0016] FIG. 1 illustrates a block diagram of an example circuit **100**. In some examples, the circuit **100** is implemented in a power converter integrated circuit (IC) or SOC module, such as a DC-DC module. The circuit **100** includes a converter circuit **102** having an output terminal **104**, and the converter circuit **102** is configured to provide an output voltage V_{out} at **104** based on an input voltage V_{in} at an input voltage terminal **106**. In some examples, the converter circuit **102** is implemented as a DC-DC converter circuit like a buck converter, boost converter etc. However, the converter circuit **102** may be implemented differently in other examples. The circuit **100** further includes a converter control circuit **108** having a first driver output terminal **110** and a second driver output terminal **112**. The converter control circuit **108** is configured to provide a high-side driving signal at the first driver output terminal **110** and a low-side driving signal at the second driver output terminal **112**. The converter control circuit **108** is configured to provide the high-side driving signal to drive a high-side switch (not shown in FIG. 1) of the converter circuit **102** and the low-side driving signal to drive a low-side switch (not shown in FIG. 1) of the converter circuit **102**. The high-side driving signal and the low-side driving signal thus control the switching of the high-side switch and the low-side switch of the converter circuit **102**, respectively.

[0017] The converter control circuit **108** is configured to generate the high-side driving signal and the low-side driving signal based on a feedback voltage V_{FB} (e.g., via a feedback loop) and an output voltage reference signal REF . In particular, the converter control circuit **108** is configured to compare the feedback voltage V_{FB} and the output voltage reference signal REF , in order to generate the high-side driving signal and the low-side driving signal. The feedback voltage V_{FB} is derived based on the output voltage V_{out} at **104**. In some examples, the feedback voltage V_{FB} is equal to the output voltage V_{out} . However, in other examples, the feedback voltage V_{FB} may be different from the output voltage V_{out} . For example, the feedback voltage V_{FB} may include a voltage divided version of the output voltage that is proportional to V_{out} . The converter control circuit **108** also has an input **114** to receive a one shot signal, shown as $SHOT$. The converter control circuit **108** is configured to control/adjust an on-time T_{on} of the high-side switch of the converter circuit **102** based on the signal $SHOT$. For example, the converter control circuit **108** can vary the high-side driving signal and low-side driving signal in accordance with the shot signal $SHOT$, and thereby control the on-time T_{on} of the high-side switch of the converter circuit **102**.

[0018] The circuit **100** further includes a one-shot circuit **132** having an output coupled to the input **114** of the converter control circuit **108**. The one-shot circuit **132** is configured to generate the shot signal $SHOT$. The one-shot circuit **132** includes a comparator **134** that includes a first comparator input **136**, a second comparator input **138** and a comparator output **140**. The comparator output **140** corresponds to the output of the one shot circuit **132**, which is coupled to the input **114** of the converter control circuit **108**. The first comparator input **136** is coupled to a voltage reference terminal and the second comparator input **138** is coupled to a compensation output **117** of a current compensation circuit **116**. The one-shot circuit **132** further includes a switch **142** coupled between the second comparator input **138** and a ground terminal. The switch **142** can be implemented as a semiconductor or other type of switching device, such as a transistor (e.g., field effect transistor (FET), bipolar junction transistor (BJT)), a thyristor or the like. The one-shot circuit **132** further includes a capacitor **144** having a first capacitor terminal **148** and a second capacitor terminal **150**, in which the first capacitor terminal **148** is coupled to the second comparator input **138** and the second capacitor terminal **150** is coupled to a ground terminal.

[0019] The capacitor **144** is configured to provide a capacitor voltage V_{cap} based on a compensated current $I_{sub.C}$. In particular, the capacitor **144** is charged based on the compensated current $I_{sub.C}$, thereby providing the capacitor voltage V_{cap} at the first capacitor terminal **148** and at the second comparator input **138**. The comparator **134** is configured to provide a comparator signal (e.g., the shot signal $SHOT$) at the comparator output **140** based on the capacitor voltage V_{cap} at the second comparator input **138** and a reference voltage V_{ref} at the first comparator input **136**. The terms comparator signal and the shot signal $SHOT$ are used interchangeably throughout

the description and are construed to be the same. The comparator signal changes states (e.g., polarity) based on a comparison of the capacitor voltage V_{cap} at the second comparator input **138** and a reference voltage V_{ref} at the first comparator input **136**. The comparator signal has a first value/polarity responsive to the capacitor voltage V_{cap} becoming greater than the reference voltage V_{ref} , and the comparator signal has a second value/polarity responsive to the capacitor voltage V_{cap} becoming lesser than the reference voltage V_{ref} . The switch **142** is turned on when the comparator signal (e.g., the shot signal SHOT) has the first value/polarity. The switch **142** is configured to discharge the capacitor **144** when the switch **142** is turned on. The turning on and turning off of the switch **142** may be controlled by providing a switching signal **115**.

[0020] The current compensation circuit **116** includes a first compensation input **118**, a second compensation input **120** and a compensation output **117**. The first compensation input **118** is coupled to the input voltage terminal **106** of the converter circuit **102** via a voltage sensor circuit **122**. The second compensation input **120** is coupled to a switching terminal **107** of the converter circuit **102** via the voltage sensor circuit **122**. The current compensation circuit **116** is configured to provide the compensated current $I_{sub.C}$ based on the input signals received at **118** and **120**.

[0021] The voltage sensor circuit **122** includes a first sensor input **128** and a first sensor output **124**, in which the first sensor input **128** is coupled to the input voltage terminal **106** and the first sensor output **124** is coupled to the first compensation input **118**. The voltage sensor circuit **122** further includes a second sensor input **130** and a second sensor output **126**, in which the second sensor input **130** is coupled to the switching terminal **107** and the second sensor output **126** is coupled to the second compensation input **120**. The voltage sensor circuit **122** is configured to provide a first voltage at the first sensor output **124** based on the input voltage V_{in} at the input voltage terminal **106**. The first voltage at **124** is proportional to the input voltage V_{in} (e.g., $V_{in}/N1$, where $N1$ is a positive value, such as an integer or fractional value). The voltage sensor circuit **122** is also configured to provide a second voltage at the second sensor output **126** based on a switching voltage V_{SW} at the switching terminal **107**. The second voltage at **126** is proportional to an average switching voltage V_{SWavg} that constitutes an average value of the switching voltage V_{SW} at **107** (e.g., $V_{SWavg}/N2$, where $N2$ is a positive value, such as an integer or fractional value). The values of $N1$ and $N2$ can be the same or different. The average switching voltage V_{SWavg} can be equal to the output voltage V_{out} , and the second voltage at the second sensor output **126** can be proportional to the output voltage V_{out} . In some examples, the voltage sensor circuit **122** includes a first voltage divider circuit and a second voltage divider circuit configured to provide the respective signals/voltages at **124** and **126**.

[0022] The current compensation circuit **116** is configured to provide the compensated current $I_{sub.C}$ based on the first voltage $V_{in}/N1$ at the first compensation input **118** and the second voltage $V_{SWavg}/N2$ at the second compensation input **120**. For example, the compensated current $I_{sub.C}$ is proportional to a difference between the input voltage V_{in} and the output voltage V_{out} . In some examples, the circuit **100** may be implemented as a system-on-chip (SOC) device in a packaging material.

[0023] FIG. 2 is a system **200** that includes the circuit **100** of FIG. 1. The system **200** provides a useful example of the converter circuit **102** and the converter control circuit **108** shown in FIG. 1. Accordingly, the description of FIG. 2 also refers to FIG. 1. In some examples, the circuit **100** is implemented as a semiconductor device (e.g., an SOC device) in a packaging material. In other examples, additional parts of the system **200** can be included in the SOC device.

[0024] The circuit **100** includes a converter circuit **102** that includes an output terminal **104** and an input voltage terminal **106**. In the example of FIG. 2, the converter circuit **102** is implemented as a DC-DC buck converter. However, in other examples, the converter circuit **102** may be implemented differently. The converter circuit **102** includes a high-side switch **202** having a first high-side terminal **206** and a second high-side terminal **208**, in which the first high-side terminal **206** is coupled to the input voltage terminal **106** and the second high-side terminal **208** is coupled

to a switching terminal **107**. The converter circuit **102** further includes a low-side switch **204** having a first low-side terminal **210** and a second low-side terminal **212**, in which the first low-side terminal **210** is coupled to the switching terminal **107** and the second low-side terminal **212** is coupled to a ground terminal. In the example of FIG. 2, the high-side switch **202** and the low-side switch **204** are implemented as FETs. In other examples, the high-side switch **202** and the low-side switch **204** can be implemented as another type of switching device, such as a BJT, a thyristor or the like.

[0025] The converter circuit **102** is configured to provide a switching voltage VSW at the switching terminal **107** between the high-side switch **202** and the low-side switch **204** based on an input voltage Vin at the input voltage terminal **106**. The converter circuit **102** further includes an inductor **216** coupled between the switching terminal **107** and the output terminal **104**. The inductor **216** includes a first inductor terminal **215** and a second inductor terminal **217**. The first inductor terminal **215** is coupled to the switching terminal **107** and the second inductor terminal **217** is coupled to the output terminal **104**.

[0026] The system **200** further includes an output capacitor **218** coupled between the output terminal **104** and a ground terminal. In some examples, the output capacitor **218** constitutes a part of the buck converter included within the converter circuit **102**. In addition, the system **200** includes a voltage divider circuit **220** configured to generate a feedback voltage VFB based on the output voltage Vout. The voltage divider circuit **220** includes a first resistor **222** and a second resistor **224** coupled in series to one another between the output terminal **104** and the ground terminal. The first resistor **222** is coupled between the output terminal **104** and the second resistor **224**. The second resistor **225** is coupled between the first resistor **222** and a ground terminal.

[0027] The circuit **100** further includes a converter control circuit **108** having a first driver output terminal **110** and a second driver output terminal **112**. The converter control circuit **108** is configured to provide a high-side driving signal at the first driver output terminal **110** to drive the high-side switch **202** of the converter circuit **102**. The converter control circuit **108** is further configured to provide a low-side driving signal at the second driver output terminal **112** to drive the low-side switch **204** of the converter circuit **102**. The converter circuit **102** is configured to provide an output voltage Vout at the output terminal **104** based on the input voltage Vin at an input voltage terminal **106**, the high-side driving signal and the low-side driving signal.

[0028] The converter control circuit **108** is configured to provide the high-side driving signal at the first driver output terminal **110** and the low-side driving signal at the second driver output terminal **112**, based on a comparator output signal COMP and a shot signal SHOT at **114**. The converter control circuit **108** in FIG. 2 includes a voltage comparator circuit **228** having a first voltage comparator input **230** and a second voltage comparator input **232**, and a voltage comparator output **234**. The first voltage comparator input **230** is coupled to the output terminal **104**. In the example of FIG. 2, the first voltage comparator input **230** is coupled to the output terminal **104** through the voltage divider circuit **220** to receive the feedback voltage VFB at the juncture between resistors **222** and **224**. The second voltage comparator input **232** is coupled to an output voltage reference terminal that provides an output voltage reference signal REF. The voltage comparator circuit **230** is further configured to compare the feedback voltage VFB and the output voltage reference signal REF to generate the comparator output signal COMP at the voltage comparator output **234**. The circuit **100** further includes a one-shot circuit **132** to provide the shot signal SHOT. The one-shot circuit **132** in FIG. 2 has the same structure of the one-shot circuit **132** in FIG. 1 and therefore the explanations are not repeated herein.

[0029] The converter control circuit **108** in FIG. 2 further includes a logic circuit **238** including a first logic input **240** and a second logic input **242**, and a first logic output **244** and a second logic output **246**. In some examples, the logic circuit **238** is implemented as an S-R flip flop. The first logic input **240** is coupled to the voltage comparator output **234** that provides the comparator output signal COMP. The second logic input **242** is coupled to a comparator output **140** of the one-

shot circuit **132**, which is configured to provide the shot signal SHOT. The second logic input **242** may be equivalent to or coupled to the input **114** of the converter control circuit **108**. The logic circuit **238** is configured to provide a first driver signal at the first logic output **244** and is configured to provide a second driver signal at the second logic output **246** based on the comparator output signal COMP and the shot signal SHOT. The first logic output **244** is coupled to an input of a high-side driver circuit **248** and the second logic output **246** is coupled to an input of a low-side driver circuit **250**. The high-side driver circuit **248** is configured to provide the high-side driving signal at the first driver output terminal **110** based on the first driver signal provided at **244**. The low-side driver circuit **250** is configured to provide the low-side driving signal at the second driver output terminal **112** based on the second driver signal provided at **246**. The high-side driving signal and the low-side driving signal are configured to control the switching of the high-side switch **202** and the low-side switch **204**, respectively, of the converter circuit **102**. In some examples, when the high-side driving signal is high, the low-side driving signal is low and vice-versa (e.g., the low-side driving signal is an inverted version of the high-side driving signal).

[0030] The one-shot circuit **132** is configured to generate the shot signal SHOT based on the reference voltage Vref at the first comparator input **136** and the capacitor voltage Vcap at the second comparator input **138**. The logic circuit **238** is configured to control/adjust an on-time Ton of the high-side switch **202** based on the signal SHOT, as described herein. The circuit **100** further includes a current compensation circuit **116** to provide a compensated current I.sub.C. In addition, the circuit **100** includes a voltage sensor circuit **122**.

[0031] FIG. 3 illustrates an example voltage sensor circuit **300**. The voltage sensor circuit **300** is a useful example of the voltage sensor circuit **122** in FIGS. 1 and 2. Accordingly, the description of FIG. 3 also refers to FIGS. 1 and 2. The voltage sensor circuit **300** includes a first voltage sensor circuit **302** including a first sensor input **128** and a first sensor output **124**. The first voltage sensor circuit **302** includes a first voltage divider circuit including a resistor R1 and a resistor R2 coupled in series to one another between the sensor input **128** and another voltage terminal, shown as AVSS. However, in other examples, the first voltage sensor circuit **302** may be implemented differently. In some examples, the AVSS can include voltage potential having a lower voltage than AVDD, such as a ground or negative potential. The first voltage sensor circuit **302** is configured to provide a first voltage at the first sensor output **124** based on an input voltage Vin (e.g., the input voltage Vin in FIG. 2) at the first sensor input **128**. The first voltage is proportional to the input voltage Vin. For example, the first voltage is $V_{in}/N1$, where N1 is a positive value. In the example of FIG. 3, the value of N1 is based on the voltage divider of R1 and R2, that is, $N1=(R1+R2)/R2$. In one example, N1 is equal to 5. However, other values of N1 can be used in other examples.

[0032] The voltage sensor circuit **300** further includes a second voltage sensor circuit **304** including a second sensor input **130** and a second sensor output **126**. The second voltage sensor circuit **304** includes a second voltage divider circuit including a combination of resistors R3, R4, R5 and R6, and capacitors C1 and C2. However, in other examples, the second voltage sensor circuit **304** may be implemented differently. The second voltage sensor circuit **304** is configured to provide a second voltage at the second sensor output **126** based on a switching voltage VSW at the second sensor input **130**. The second voltage sensor circuit **304** is also coupled to AVSS. The second voltage is proportional to an average switching voltage VSWavg that constitutes an average value of the switching voltage VSW. For example, the second voltage is $VSW_{avg}/N2$, where N2 is integer positive value. The resistor R5 and the capacitor C1 form a first RC filter stage, and the resistor R6 and the capacitor C2 form a second RC filter stage. In other examples, there can be more than two RC filter stages. The first RC filter stage and the second RC filter stage are configured to generate $VSW_{avg}/N2$ based on the voltage at the node between the resistors R3 and R4. The average switching voltage VSWavg can approximate or be equal to an output voltage Vout. For example, during the operation of the converter circuit (e.g., the converter circuit **102** in FIGS. 1 and 2), the switching voltage VSW switches between Vin and AVSS (or ground) based on a duty

cycle D of the converter circuit, while Vout is also generated based on Vin and the duty cycle D of the converter circuit, such that VSWavg can be equal to Vout. In the example of FIG. 3, the value of N2 is based on the voltage divider of R3 and R4, that is, $N2 = (R3 + R4) / R4$. In one example, N2 is equal to 5. Other values of N2 can also be used in other examples. In some examples, the values of the resistors R1, R2, R3 and R4 are chosen so $N1 = N2$. In other examples, different values of N1 and N2 can be used to apply different scaling for Vin and VSW.

[0033] FIG. 4 illustrates an example system 400 that includes a current compensation circuit 402 and a one-shot circuit 404. The current compensation circuit 402 provides a useful example that can be used as the current compensation circuit 116 in FIGS. 1 and 2. The current compensation circuit 402 includes a first voltage-to-current converter 406 including a first voltage input 408 and a first current output 410. The first voltage input 408 corresponds to the first compensation input 118 in FIGS. 1 and 2 and therefore receives the first voltage $V_{in}/N1$ (e.g., from first voltage sensor circuit 302). The first voltage to current converter 406 further includes a first compensation switch M1 having a first compensation terminal, a second compensation terminal and a third compensation terminal, in which the first compensation terminal constitutes the first current output 410 and the second compensation terminal constitutes the first voltage input 408. The first voltage to current converter 406 further includes a first compensation resistor R7 coupled between the third compensation terminal of the first compensation switch M1 and another voltage terminal, shown as AVSS. In some examples, the AVSS can include a ground potential or a negative potential.

[0034] The first voltage to current converter 406 is configured to provide a first current I1 at the first current output 410 based on the first voltage $V_{in}/N1$ at the first voltage input 408. For example, the first current I1 at the first current output 410 can be given as:

$$[00001] \ I_1 = \frac{V_{in}}{N1 * R7} \quad (1)$$

[0035] The current compensation circuit 402 further includes a second voltage-to-current converter 416 including a second voltage input 418 and a second current output 420. The second voltage input 418 corresponds to the second compensation input 120 in FIGS. 1 and 2 and therefore receives the second voltage $V_{SWavg}/N2$ (e.g., from second voltage sensor circuit 304). The second voltage $V_{SWavg}/N2$ can be equal to $V_{out}/N2$, in some examples. The second voltage to current converter 416 includes a second compensation switch M2 including a fourth compensation terminal, a fifth compensation terminal and a sixth compensation terminal, in which the fourth compensation terminal constitutes the second current output 420 and the fifth compensation terminal constitutes the second voltage input 418. The second voltage to current converter 416 further includes a second compensation resistor R8 coupled between the sixth compensation terminal of the second compensation switch M2 and the voltage terminal AVSS.

[0036] The second voltage to current converter 416 is configured to provide a second current I2 at the second current output 420 based on the second voltage $V_{SWavg}/N2$ at the second voltage input 418. For example, the second current I2 can be given as:

$$[00002] \ I_2 = \frac{V_{SWavg}}{N2 * R8} \quad (2)$$

[0037] The current compensation circuit 402 further includes a first current mirror network 426 having a first current mirror input 428 and a first current mirror output 430, in which the first current mirror input 428 is coupled to the first current output 410. The first current mirror network 426 is configured to mirror the first current I1 at the first current mirror input 428 to provide the first current I1 (or another current proportional to the first current I1) at the first current mirror output 430. The current compensation circuit 402 further includes a second current mirror network 432 including a second current mirror input 434 and a second current mirror output 436, in which the second current mirror input 434 is coupled to second current output 420. The second current mirror output 436 constitutes a compensation output (e.g., the compensation output 117 in FIGS. 1 and 2) of the current compensation circuit 402. The second current mirror network 432 includes a first current mirror 448 and a second current mirror 450 that are coupled to one another. The first

current mirror network **426** and the second current mirror network **432** are coupled to a voltage terminal, shown as AVDD. In some example, the voltage terminal AVDD includes a positive voltage terminal, such as can be coupled to an output of a DC voltage source.

[0038] The second current mirror input **434** is configured to receive a current I₃, which is equal to the difference between I₁ and I₂. For example, the current at the second current mirror input **434** includes a third current I₃ given as:

[00003] $I_3 = I_1 - I_2$ (3)

[0039] Substituting equation (1) and (2) in equation (3) and assuming that N=N₁=N₂, provides

[00004] $I_3 = \frac{V_{in}}{N \cdot R_7} - \frac{V_{sWavg}}{N \cdot R_8}$ (4)

[0040] Considering R₇=R₈=R, and V_SWavg=V_{out}, equation (3) becomes:

[00005] $I_3 = \frac{1}{NR}(V_{in} - V_{out})$ (5)

[0041] The second current mirror network **432** is configured to mirror the third current I₃ at the second current mirror input **434** to provide a compensated current I_{sub.C} at the second current mirror output **436**. In some examples, the compensated current I_{sub.C} at the second current mirror output **436** is equal to the third current I₃. However, in other examples, the compensated current I_{sub.C} at the second current mirror output **436** may include a current that is proportional to the third current I₃ according to the configuration of the current mirror network **432**. The compensated current I_{sub.C} at the second current mirror output **436** corresponds to the compensated current I_{sub.C} in FIGS. **1** and **2**. The compensated current I_{sub.C} can be given as:

[00006] $I_C = \frac{1}{NR}(V_{in} - V_{out})$ (6)

[0042] The one-shot circuit **404** includes a comparator **440** that includes a first comparator input **442**, a second comparator input **444** and a comparator output **441**. The first comparator input **442** is configured to receive a reference voltage V_{ref} and the second comparator input **444** is coupled to the compensation output **436**. The one-shot circuit **404** further includes a switch **446** coupled between the second comparator input **444** and the ground terminal. In the example of FIG. **4**, the switch **446** includes a FET with its gate terminal configured to receive a switching signal (e.g., the switching signal **115** in FIGS. **1** and **2**). The one-shot circuit **404** further includes a capacitor C which is configured to provide a capacitor voltage V_{cap} based on the compensated current I_{sub.C} at the second current mirror output **436**. The one-shot circuit **404** functions similarly to and provides a useful example that can be implemented as the one-shot circuit **132** in FIGS. **1** and **2**.

[0043] Referring back to FIG. **2**, the one-shot circuit **132** (e.g., circuit **404**) is configured to control/adjust an on-time T_{on} of a high-side switch **202** via the shot signal SHOT provided at **114**. In particular, the one-shot circuit **132** is configured to control the turning off of the high-side switch **202** based on the shot signal SHOT at **114**. The on-time T_{on} of the high-side switch **202** corresponds to a continuous time period during which the high-side switch **202** is turned on. In order to facilitate the one-shot circuit **132** to adjust the on-time T_{on} of the high-side switch **202**, when the high-side switch **202** is turned on, the switch **142** of the one-shot circuit **132** is turned off. Conversely, when the high-side switch **202** is turned off, the switch **142** of the one-shot circuit **132** is turned on. The turning on and turning off of the switch **142** may be controlled by providing a switching signal **115**. In some examples, the switching signal **115** includes an inverted version of the high-side driving signal of the high-side switch **202**. However, in other examples, the switching signal **115** may be implemented differently, such as the low-side driving signal at **112**.

[0044] The logic circuit **238** is configured to control turning on of the high-side switch **202** based on the comparator output signal COMP and the SHOT signal. For example, when the feedback voltage V_{FB} at the first voltage comparator input **230** becomes lesser than the output voltage reference REF at the second voltage comparator input **232**, the comparator output signal COMP goes low, thereby turning on the high-side switch **202**. To better understand the turning on and the turning off of the high-side switch **202**, FIG. **5** illustrates a signal diagram **500** that depicts the

comparator output signal COMP, the shot signal SHOT and the switching voltage VSW of FIG. 2. As can be seen in FIG. 5, when the comparator output signal COMP goes low at T1, the switching voltage VSW goes high, which in turn indicates that the high-side switch **202** is turned on.

[0045] Referring back to FIG. 2, when the high-side switch **202** is turned on, the switch **142** of the one-shot circuit **132** is turned off, thereby charging the capacitor **144**, based on the compensated current I.sub.C. When the voltage Vcap across the capacitor **144** becomes equal to or greater than Vref, the comparator signal changes states (i.e., the shot signal SHOT goes low), thereby turning off the high-side switch **202**. Referring to FIG. 5, when the shot signal SHOT goes low at T2, the switching voltage VSW goes low, which in turn indicates that the high-side switch **202** is turned off. Referring back to FIG. 2, when the high-side switch **202** is turned off, the switch **142** of the one-shot circuit **132** turns on, thereby discharging the capacitor **144**. Once the high-side switch **202** is turned off, the capacitor **144** waits to be charged until the next time the high-side switch **202** is turned on.

[0046] Therefore, based on the example arrangement shown in FIG. 2, the compensated current I.sub.C charges the capacitor **144** to a voltage Vref (e.g., from 0V) during the on-time Ton (e.g., the time period T2–T1 in FIG. 5) of the high-side switch **202**. For example, the on-time Ton of the high-side switch **202** can be derived as follows:

[00007] $I_C = C \frac{dv}{dt}$ (7) [0047] where: I.sub.C=the compensated current; [0048] C=the capacitance value of the capacitor **144**; [0049] dt=Ton; and [0050] dv=Vref-0=Vref (that is, the change in capacitor voltage during Ton)

[0051] Substituting the above values in equation (7), the current value becomes:

[00008] $I_C = C \frac{V_{ref}}{T_{on}}$ (8)

[0052] Further substituting equation (6) in equation (8), the value of the on-time Ton can be represented as follows:

[00009] $\frac{1}{NR}(V_{in} - V_{out}) = C \frac{V_{ref}}{T_{on}}$ (9)

Therefore, $T_{on} = \frac{N * R * C * V_{ref}}{(V_{in} - V_{out})}$ (10)

[0053] As described herein, controlling the on-time Ton of the high-side switch **202** based on the shot signal SHOT enables the circuit **100** to have a constant inductor ripple current across the inductor **216** of the converter circuit **102**. In particular, controlling the on-time Ton of the high-side switch **202** based on the shot signal SHOT enables a current through the inductor **216** to have a constant inductor ripple current. For example, voltage across the inductor **216** is given as:

[00010] $V_L = L \frac{dI_L}{dt}$ (11)

[0054] During the on-time Ton of the high-side switch **202**, dI.sub.L=Ipp which is the inductor ripple current, dt=Ton and V.sub.L=Vin-Vout (as the switching voltage VSW becomes equal to Vin when the high-side switch **202** is turned on).

[0055] Substituting the above values in equation (11), the equation for inductor ripple current (through the inductor **216**) can be expressed as follows:

[00011] $I_{pp} = \frac{(V_{in} - V_{out})}{L} * T_{on}$ (12)

[0056] Where Ipp is a peak to peak inductor ripple current.

[0057] Substituting for Ton based on equation (10), the inductor ripple current Ipp can be expressed as follows:

[00012] $I_{pp} = \frac{N * R * C * V_{ref}}{L}$ (13)

[0058] Thus, as can be seen from equation (13), inductor ripple current Ipp depends on all constant factors, which can result in a constant inductor ripple current Ipp across the inductor **216**.

Advantageously, the inductor ripple current Ipp is independent of the output voltage Vout, and therefore, the inductor ripple current is constant irrespective of the value of the output voltage Vout.

[0059] As a result, the current compensation circuit **116** and the one-shot circuit **132** as described

herein can be utilized to implement power converter modules (e.g., DC-DC converter SOC modules) over a wide range of output voltages without affecting the inductor ripple current. For example, in power converter modules with a fixed integrated inductor (e.g., inductor **216**), both high and low values of output voltage V_{out} can be supported without a change in the inductor ripple current. Further, this approach enables the use of smaller inductors to achieve a certain voltage level in power converter modules, thereby reducing the overall size and cost of the power converter modules.

[0060] In this description, the term “based on” means based at least in part on.

[0061] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0062] Also, in this description, a device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

Furthermore, a circuit or device described herein as including certain components may instead be configured to couple to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be configured to couple to at least some of the passive elements and/or the sources to form the described structure, either at a time of manufacture or after a time of manufacture, such as by an end user and/or a third party.

[0063] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A circuit comprising: a current compensation circuit having first and second voltage input terminals and a current output terminal; and a one-shot circuit including: a capacitor having first and second terminals, the first terminal coupled to the current output terminal, the second terminal coupled to the current compensation circuit; and a comparator having a reference input and a second input, the second input coupled to the first terminal of the capacitor and the current output terminal of the current compensation circuit.
2. The circuit of claim 1, wherein the one-shot circuit includes a switch having first and second terminals, the first terminal of the switch coupled to the first terminal of the capacitor, the second terminal of the switch coupled to the second terminal of the capacitor.
3. The circuit of claim 1, wherein the current compensation circuit includes: a first transistor having first and second terminals and a control terminal, the second terminal coupled to the second terminal of the capacitor, the control terminal coupled to the first voltage input terminal of the current compensation circuit; a second transistor having first and second terminals and a control terminal, the first terminal coupled to the current output terminal of the current compensation circuit, the second terminal coupled to second terminal of the capacitor, the control terminal

coupled to the second voltage input terminal of the current compensation circuit; and a current mirror having a first terminal coupled to the first terminal of the first transistor and having a second terminal coupled to the first terminal of the second transistor.

4. The circuit of claim 3, wherein the current compensation circuit includes current mirror circuitry having a first terminal coupled to the first terminal of the second transistor and having a second terminal coupled to the current output terminal of the current compensation circuit.

5. The circuit of claim 1 further comprising: a first transistor having first and second terminals, the first terminal coupled to a third voltage input terminal, which is coupled to the first voltage input terminal of the current compensation circuit, the second terminal of the first transistor coupled to a switching terminal, which is coupled to the second voltage input terminal of the current compensation circuit; and a second transistor having first and second terminals, the first terminal coupled to the switching terminal, the second terminal coupled to a fourth voltage input terminal.

6. The circuit of claim 5, wherein the current compensation circuit is capable of providing a current at the current output terminal that is based on a first voltage at the third input voltage terminal and a second voltage at the switching terminal.

7. The circuit of claim 6, wherein the comparator is capable of providing a signal to turn off the first transistor responsive to the current and a voltage at the first terminal of the capacitor exceeding a reference voltage at the reference terminal of the comparator.

8. The circuit of claim 1, wherein the current compensation circuit is capable of providing a current at the current output terminal that is based on an input voltage at a terminal of a high-side transistor of a converter circuit and based on a voltage at a switching terminal of the converter.

9. A circuit comprising: a first transistor having first and second terminals and a control terminal, the first terminal coupled to an input voltage terminal, the second terminal coupled to a switching terminal; a second transistor having a terminal coupled to the switching terminal, the; an inductor having a terminal coupled to the switching terminal; a voltage sensor circuit having a first input coupled to the input voltage terminal and a second input coupled to the switching input, and having first and second outputs; a current compensation circuit having a first input coupled to the first output of the voltage sensor circuit, a second input coupled to the second output of the voltage sensor circuit, and an output; and a one-shot circuit including: a capacitor having first and second terminals, the first terminal coupled to the output of the current compensation circuit; a comparator having a reference input, a second input, and an output, the second input coupled to the first terminal of the capacitor and the output of the current compensation circuit, the output of the comparator coupled to the control terminal of the first transistor; and a switch having first and second terminals, the first terminal of the switch coupled to the first terminal of the capacitor, the second terminal of the switch coupled to the second terminal of the capacitor.

10. The circuit of claim 9, wherein the current compensation circuit includes: a third transistor having first and second terminals and a control terminal, the control terminal coupled to the first input of the current compensation circuit; a fourth transistor having first and second terminals and a control terminal, the first terminal coupled to the output of the current compensation circuit, the second terminal coupled to second terminal of the third transistor, the control terminal coupled to the second input of the current compensation circuit; and a current mirror having a first terminal coupled to the first terminal of the third transistor and having a second terminal coupled to the first terminal of the fourth transistor.

11. The circuit of claim 10, wherein the current compensation circuit includes current mirror circuitry having a first terminal coupled to the first terminal of the fourth transistor and having a second terminal coupled to the output of the current compensation circuit.

12. The circuit of claim 9, wherein the circuit is a system-on-chip device in a packaging material.

13. The circuit of claim 9, wherein the voltage sensor circuit includes: a first voltage divider circuit coupled to the first input and the first output of the voltage sensor circuit; and a second voltage divider circuit coupled to the second input and the second output of the voltage sensor circuit.

14. The circuit of claim 9, wherein the comparator is a first comparator, and the circuit further comprising a control circuit including: a logic circuit having an input coupled to the output of the comparator, and first and second outputs; a first driver having an input coupled to the first output of the logic circuit and an output coupled to the control terminal of the first transistor; a second driver having an input coupled to the second output of the logic circuit and an output coupled to a control terminal of the second transistor; and a second comparator having a feedback input and a reference input.

15. A circuit comprising: a transistor having first and second terminals and a gate, the first terminal coupled to an input voltage terminal, the second terminal coupled to a switching terminal; a current compensation circuit having a first input coupled to the input voltage terminal, a second input coupled to the switching terminal, and an output; a capacitor having a terminal coupled to the output of the current compensation circuit; a first comparator having a reference input, a second input, and an output, the second input coupled to the terminal of the capacitor and the output of the current compensation circuit; a second comparator having a feedback input, a reference input, and an output; a logic circuit having first and second inputs and an output, the first input coupled to the output of the first comparator, the second input coupled to the output of the second comparator; and a gate driver having an input coupled to the output of the logic circuit and an output coupled to the gate of the transistor.

16. The circuit of claim 15, wherein the circuit is a system-on-chip device in a packaging material, the transistor is a first transistor, the gate driver is a first gate driver, and the circuit further comprising: an inductor having a terminal coupled to the switching terminal; a second transistor having a terminal coupled to the switching terminal, and a gate; and a second gate driver having an input coupled to a second output of the logic circuit, and an output coupled to the gate of the second transistor.

17. The circuit of claim 15, further comprising: a first voltage divider circuit coupled to the input voltage terminal and the first input of the current compensation circuit; and a second voltage divider circuit coupled to the switching terminal and the second input of the current compensation circuit.

18. The circuit of claim 15, wherein the current compensation circuit is capable of providing a current at its output based on a first voltage at the input voltage terminal and a second voltage at the switching.

19. The circuit of claim 15, wherein the current compensation circuit capable of providing a current at its output based on a first voltage and a second voltage, in which the first voltage is proportion to a voltage at the input voltage terminal and the second voltage is proportional to a voltage at the switching terminal.

20. The circuit of claim 15, wherein, responsive to the current, the comparator is capable of providing a signal at its output to control the logic circuit to turn the transistor off.
